

XUANYU WANG

wang.xuanyu@icloud.com | xuanyuwang.com |  xuanyu-wang

SUMMARY

Senior backend engineer with 7+ years building data-intensive product systems. Diagnoses cross-system correctness failures, turns ambiguous semantics into explicit contracts, and owns solutions through architecture, safe rollout, and production verification.

EXPERIENCE

Software Engineer
Cresta - Conversation Intelligence (Coaching, QM, Analytics)

Dec. 2023 - Present
Toronto, Canada

- Designed and globally rolled out a general ClickHouse external-table path for large analytics filters; evaluated 10 approaches, integrated generic helpers across 17 caller files, and benchmarked 3.3× faster queries at 10,000 users.
- Diagnosed PostgreSQL-to-ClickHouse scorecard inconsistencies caused by asynchronous stale writes and ORM lost updates; shipped atomic, post-commit re-read and partial-update fixes with zero score or submitter mismatches across 2,996 comparable production records over 39 days.
- Built fleet-wide scorecard sync monitoring and safety-gated repair across seven production clusters, classifying missing and stale records and dispatching targeted ID-based reindex jobs with allowlists, thresholds, and workflow chunking.
- Defined user-filter semantics across three divergent implementations; created an implementation-independent behavioral standard and 62-test suite, identified silent ACL and selection differences, and fixed a production union-versus-intersection defect.
- Led backend architecture and delivery for Group Calibration, including task APIs, response workflows, consistency scoring, analytics, and notifications for end-to-end quality-management calibration.

Software Engineer (Data)
Boosted.ai - AI-driven investment platform

2021 - 2023
Toronto, Canada

- Designed and maintained ETL pipelines ingesting global financial market data across multiple exchanges, enabling daily analytics and model-driven investment workflows.
- Built a high-performance data access service with FastAPI, PostgreSQL, and Solr, providing internal teams a unified interface for large financial datasets.
- Developed gRPC microservices for user-data management and investment-signal generation, supporting personalization workflows.
- Prototyped and advocated Prefect-based orchestration to improve pipeline reliability and operational visibility.
- Standardized Docker-based development environments, reducing local setup from days to minutes.

Software Developer (Frontend), Scrum Master
IBM - Cognos Analytics

2018 - 2021
Ottawa, Canada

- Developed and shipped React and TypeScript features for a large-scale analytics platform.
- Improved Crosstab rendering performance by simplifying DOM structure by more than 50%, improving responsiveness for large datasets.

SKILLS

Backend: Go, Python
Data Systems: PostgreSQL, ClickHouse
Frontend: React, TypeScript